

MATH 394 B, Joint with STAT 394 B
Probability I
Course type: Unknown

Evaluation Delivery: Online
Evaluation Form: A
Responses: 14/21 (67% high)

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Predoc Inst

Overall Summative Rating represents the combined responses of students to the four global summative items and is presented to provide an overall index of the class's quality:

Combined Median	Adjusted Combined Median
4.1	4.1
(0=lowest; 5=highest)	

Challenge and Engagement Index (CEI) combines student responses to several *IASystem* items relating to how academically challenging students found the course to be and how engaged they were:

CEI: 5.0
(1=lowest; 7=highest)

SUMMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Adjusted Median
The course as a whole was:	14	29%	36%	21%	14%			3.9	3.9
The course content was:	14	29%	43%	29%				4.0	4.0
The instructor's contribution to the course was:	14	50%	29%	14%	7%			4.5	4.5
The instructor's effectiveness in teaching the subject matter was:	14	29%	43%	14%	7%	7%		4.0	4.0

STUDENT ENGAGEMENT

Relative to other college courses you have taken:	N	Much Higher (7)	(6)	(5)	Average (4)	(3)	(2)	Much Lower (1)	Median
Do you expect your grade in this course to be:	13	8%	31%	15%	38%	8%			4.8
The intellectual challenge presented was:	13	8%	38%	31%	15%	8%			5.4
The amount of effort you put into this course was:	13	15%	31%	31%	23%				5.4
The amount of effort to succeed in this course was:	13	23%	23%	31%	23%				5.4
Your involvement in course (doing assignments, attending classes, etc.) was:	13	15%	23%	23%	38%				5.0

On average, how many hours per week have you spent on this course, including attending classes, doing readings, reviewing notes, writing papers and any other course related work?

Class median: 10.0 Hours per credit: 2.5 (N=11)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
		9%	9%	27%	18%	18%	9%	9%			

From the total average hours above, how many do you consider were valuable in advancing your education?

Class median: 8.5 Hours per credit: 2.1 (N=11)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
9%	18%	9%	9%	9%	27%	9%	9%				

What grade do you expect in this course?

Class median: 3.5 (N=11)

A (3.9-4.0)	A- (3.5-3.8)	B+ (3.2-3.4)	B (2.9-3.1)	B- (2.5-2.8)	C+ (2.2-2.4)	C (1.9-2.1)	C- (1.5-1.8)	D+ (1.2-1.4)	D (0.9-1.1)	D- (0.7-0.8)	F (0.0)	Pass	Credit	No Credit
36%	18%	18%	9%	9%	9%									

In regard to your academic program, is this course best described as:

(N=12)

In your major	A core/distribution requirement	An elective	In your minor	A program requirement	Other
33%	8%	17%	17%	17%	8%

STANDARD FORMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Relative Rank
Course organization was:	13	23%	38%	15%	23%			3.8	15
Clarity of instructor's voice was:	13	54%	23%	23%				4.6	8
Explanations by instructor were:	13	38%	38%	8%	8%	8%		4.2	10
Instructor's ability to present alternative explanations when needed was:	12	33%	25%	33%	8%			3.8	18
Instructor's use of examples and illustrations was:	13	62%	23%	8%	8%			4.7	1
Quality of questions or problems raised by the instructor was:	12	33%	33%	17%	17%			4.0	16
Student confidence in instructor's knowledge was:	12	50%	33%	17%				4.5	9
Instructor's enthusiasm was:	12	42%	42%		17%			4.3	14
Encouragement given students to express themselves was:	11	55%	36%			9%		4.6	6
Answers to student questions were:	12	50%	25%	8%	17%			4.5	5
Availability of extra help when needed was:	12	50%	33%	8%	8%			4.5	7
Use of class time was:	12	33%	42%	17%		8%		4.1	12
Instructor's interest in whether students learned was:	12	42%	50%		8%			4.3	11
Amount you learned in the course was:	12	33%	25%	8%	25%	8%		3.8	17
Relevance and usefulness of course content were:	12	50%	17%	17%	17%			4.5	3
Evaluative and grading techniques (tests, papers, projects, etc.) were:	12	58%	33%	8%				4.6	2
Reasonableness of assigned work was:	12	42%	25%	25%	8%			4.2	13
Clarity of student responsibilities and requirements was:	12	50%	25%	17%	8%			4.5	4

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STANDARD OPEN-ENDED QUESTIONS

Was this class intellectually stimulating? Did it stretch your thinking? Why or why not?

1. It was. The content itself touches many branches of math and connects them, which was my favorite part of this course.
2. Yes, this class was intellectually stimulating. It helped me think more carefully about probability and solve problems in different ways.
3. Yes, as an economics major this class has been extremely helpful. I was able to enhance my understanding in the content I learned in my economics class by learning more mathematical sides to it.
4. Class taught me a lot of subject of math i'm not very familiar with
7. I found this class very interesting!
8. absolutely. even for a math nerd there were some good concepts

What aspects of this class contributed most to your learning?

1. The homework assignments really helped to cement my understanding of the material.
2. The examples, practice problems, and clear explanations contributed the most to my learning.
3. I was able to learn about different distributions (in economics I had only learned normal distribution) and a deeper analysis regarding statistics.
4. The course website was very well made and organized and all the materials prepared for exam prep were very useful.
5. I liked having the slides to look back on, and the updated slides with instructor notes. Having homework solutions posted and recorded lectures was so good for looking back on information and learning from my mistakes, I wish all math classes did that.
6. Having slides posted and annotated slides posted, and having zoom recordings available to review after lectures whenever needed
7. I liked that Arman made visualizations of some of the concepts we were learning. Seeing a visual representation of the concepts helped a lot with my understanding.
8. good homework problems.

What aspects of this class detracted from your learning?

1. I felt that every so often there would be a homework question that was unreasonably hard and take hours upon hours to solve.
2. Sometimes the class moved a little too quickly, so some difficult topics were hard to understand.
3. N/A
4. Me being very unfamiliar with a lot of the notation used in the class hurt my learning experience. It also felt there was a substantial gap between the hw and content taught in class.
5. The way the slides were organized sometimes felt confusing. If I got lost in lecture, it wasn't always clear exactly what concept we were working on. Might have been my issue for getting lost in lecture. Sometimes the homework assignments did not feel related enough to what we covered in lecture. I think more examples might help with this. Sometimes I googled questions because I did not feel like I had a strong enough understanding of how a concept applied to a situation.
7. I sometimes the lectures and the homeworks were pretty different. More examples that connect to the homework in later lectures would've been better.
8. none

What suggestions do you have for improving the class?

1. Other than some of the homework problems being too hard and time consuming, I thought this was an excellent class with little to improve on.
2. I suggest spending more time on difficult topics and providing more practice problems before exams.
3. Keep this up. He did an excellent job in explaining the concepts and creating homeworks and exams.
4. More examples related to the hw in class would be nice
5. The homework assignments for a certain week's content should be due the week after we learn about it. Even though that is a big break from the day the content is taught to the day the homework is due, I feel it is worse to be working on Sunday's homework assignment on Wednesday and feeling like I don't know enough to get even half of it done.
8. keep leaning into those homework problems and the proofs, i really appreciated them.

IASystem Course Summary Reports summarize student ratings of a particular course or combination of courses. They provide a rich perspective on student views by reporting responses in three ways: as frequency distributions, average ratings, and either comparative or adjusted ratings. Remember in interpreting results that it is important to keep in mind the number of students who evaluated the course relative to the total course enrollment as shown on the upper right-hand corner of the report.

Frequency distributions. The percentage of students who selected each response choice is displayed for each item. Percentages are based on the number of students who answered the respective item rather than the number of students who evaluated the course because individual item response is optional.

Median ratings. IASystem reports average ratings in the form of item medians. Although means are a more familiar type of average than medians, they are less accurate in summarizing student ratings. This is because ratings distributions tend to be strongly skewed. That is, most of the ratings are at the high end of the scale and trail off to the low end.

The median indicates the point on the rating scale at which half of the students selected higher ratings, and half selected lower. Medians are computed to one decimal place by interpolation.¹ In general, higher medians reflect more favorable ratings. To interpret median ratings, compare the value of each median to the respective response scale: *Very Poor, Poor, Fair, Good, Very Good, Excellent (0-5); Never/None/Much Lower, About Half/Average, Always/Great/Much Higher (1-7); Slight, Moderate, Considerable, Extensive (1-4)*.

Comparative ratings. IASystem provides a normative comparison for each item by reporting the decile rank of the item median. Decile ranks compare the median rating of a particular item to ratings of the same item over the previous two academic years in all classes at the institution and within the college, school, or division. Decile ranks are shown only for items with sufficient normative data.

Decile ranks range from 0 (lowest) to 9 (highest). For all items, higher medians yield higher decile ranks. The 0 decile rank indicates an item median in the lowest 10% of all scores. A decile rank of 1 indicates a median above the bottom 10% and below the top 80%. A decile rank of 9 indicates a median in the top 10% of all scores. Because average ratings tend to be high, a rating of "good" or "average" may have a low decile rank.

Adjusted ratings. Research has shown that student ratings may be somewhat influenced by factors such as class size, expected grade, and reason for enrollment. To correct for this, IASystem reports **adjusted medians** for summative items (items #1-4 and their combined global rating) based on regression analyses of ratings over the previous two academic years in all classes at the respective institution. If large classes at the institution tend to be rated lower than small classes, for example, the adjusted medians for large classes will be slightly higher than their unadjusted medians.

When adjusted ratings are displayed for summative items, **relative rank** is displayed for the more specific (formative) items. Rankings serve as a guide in directing instructional improvement efforts. The top ranked items (1, 2, 3, etc.) represent areas that are going well from a student perspective; whereas the bottom ranked items (18, 17, 16, etc.) represent areas in which the instructor may want to make changes. Relative ranks are computed by first standardizing each item (subtracting the overall institutional average from the item rating for the particular course, then dividing by the standard deviation of the ratings across all courses) and then ranking those standardized scores.

Challenge and Engagement Index (CEI). Several IASystem items ask students how academically challenging they found the course to be. IASystem calculates the average of these items and reports them as a single index. *The Challenge and Engagement Index (CEI)* correlates only modestly with the global rating (median of items 1-4).

Optional Items. Student responses to instructor-supplied items are summarized at the end of the evaluation report. Median responses should be interpreted in light of the specific item text and response scale used (response values 1-6 on paper evaluation forms).

¹ For the specific method, see, for example, Guilford, J.P. (1965). *Fundamental statistics in psychology and education*. New York: McGraw-Hill Book Company, pp. 49-53.